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Recovering Instruction Following Loss in Quantized LLMs via Attention Head Compensation

 (/pdf?id=GfTGioqEma)

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 06 Jan 2026 (modified: 25 Feb 2026)  ACL ARR 2026 January Submission

 January, Senior Area Chairs, Area Chairs, Reviewers, Authors

 Revisions (/revisions?id=GfTGioqEma)  BibTeX

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Keywords: feature attribution, probing

Abstract:

Model quantization is essential for deploying large language models (LLMs). However, quantized models often exhibit unpredictable failures in instruction following, including unintended language switching, violation of formatting constraints, and degenerative generation. We present \textbf{Deep Attention Stimulation (DAS)}, a training-free intervention that selectively compensates attention heads most disrupted by quantization. Inspired by targeted neural stimulation in cognitive neuroscience, DAS identifies critical attention heads by analyzing activation differences between full-precision and quantized models on instruction-following failure cases. Through qualitative analysis on 10 carefully selected samples from IFEval, we show that injecting small corrective signals into these heads can recover instruction-following behavior. In particular, a 4-bit GPTQ Qwen2.5-7B-Instruct model recovers correct English output and avoids repetitive degeneration under moderate stimulation. Our pilot study suggests that quantization-induced instruction-following failures are localized to a small subset of attention heads rather than uniformly distributed. These findings highlight the potential of interpretable and targeted post-quantization repair mechanisms.

Paper Type: Long

Research Area: Interpretability and Analysis of Models for NLP

Research Area Keywords: Interpretability and Analysis of Models for NLP, Efficient/Low-Resource Methods for NLP

Contribution Types: Model analysis & interpretability

Languages Studied: English

TLDR:  We show that Instruction-Following failures induced by low-bit quantization are highly localized to a small subset of attention heads, and introduce a training-free attention head compensation method that can recover instruction adherence in quantized LLMs

Reassignment Request Area Chair:  This is not a resubmission

Reassignment Request Reviewers:  This is not a resubmission

Preprint:  yes

Preprint Status:  There is no non-anonymous preprint and we do not intend to release one. (this option is binding)

Preferred Venue:  ACL

Consent To Share Data:  yes

Consent To Share Submission Details:  On behalf of all authors, we agree to the terms above to share our submission details.

A1 Limitations Section:  This paper has a limitations section.

A2 Potential Risks: 👁 N/A

B Use Or Create Scientific Artifacts: 👁 Yes

B4 Data Contains Personally Identifying Info Or Offensive Content: 👁 N/A

B6 Statistics For Data: 👁 Yes

B6 Elaboration: 👁 4

C Computational Experiments: 👁 Yes

C2 Experimental Setup And Hyperparameters: 👁 Yes

C2 Elaboration: 👁 4

C3 Descriptive Statistics: 👁 Yes

C3 Elaboration: 👁 5

D Human Subjects Including Annotators: 👁 No

D1 Instructions Given To Participants: 👁 N/A

D2 Recruitment And Payment: 👁 N/A

D3 Data Consent: 👁 N/A

D4 Ethics Review Board Approval: 👁 N/A

E Ai Assistants In Research Or Writing: 👁 Yes

E1 Information About Use Of Ai Assistants: 👁 Yes

E1 Elaboration: 👁 AI assistants were used to support writing and editing, including improving clarity and organization of the manuscript and assisting with the completion for the submission checklist. No AI system was used to generate experimental results, data, or scientific claims.

Author Submission Checklist: 👁 yes

Association For Computational Linguistics - Blind Submission License Agreement: 👁 On behalf of all authors, I agree

Submission Number: 7892

Discussion (?id=GfTGioqEma#discussion)

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Meta Review of Submission7892 by Area Chair B4QE

🔗

Meta Review

by Area Chair B4QE

📅 05 Mar 2026, 08:01 (modified: 11 Mar 2026, 15:59)

👁 Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs

📄 Revisions (/revisions?id=9mKrCrXf41)

Metareview:

This paper studies a practically important failure mode: noticeably degrading in a few instruction-following examples. The authors propose Deep Attention Stimulation (DAS), a training-free, head-level intervention. DAS first diagnoses “degraded heads” by comparing attention behaviors between an FP16 reference model and a GPTQ 4-bit model, focusing on instruction-bearing tokens, and then manually repairs the quantized model by injecting a scaled correction signal into selected attention-head outputs at inference time (using a stimulation strength parameter α). Experiments are presented primarily on Qwen2.5-7B-Instruct under GPTQ 4-bit, using IFEval-derived failure cases, with a qualitative emphasis on a curated set of representative examples and analysis of non-monotonic α effects.

Summary Of Reasons To Publish:

The paper addresses a timely and practically relevant question: quantization is central for deployment, yet its impact on instruction-following quality is often under-measured by standard metrics. The work makes this gap explicit and highlights concrete failure types that matter in real usage.

The head-level adjustment "diagnose to intervene" framing is conceptually interesting. Rather than treating quantization damage as diffuse, the paper attempts a mechanistic angle by localizing perturbations to specific attention components and testing targeted interventions.

The work explores a training-free post-hoc repair direction and openly reports instability regimes (e.g., over-stimulation leading to degeneration).

Summary Of Suggested Revisions:

This paper investigates the degradation of instruction-following capabilities in quantized large language models and proposes Deep Attention Stimulation (DAS), a method that identifies and compensates for attention heads most affected by quantization. While the problem is practically relevant and the interpretability-inspired intervention is conceptually interesting, the work remains at a preliminary stage and suffers from several critical limitations that undermine its main claims. First, the proposed DAS method is highly input-specific: the corrective signals are derived from activation differences between the full-precision and quantized models on specific prompts and are evaluated only on the same prompts. This raises serious concerns about generalization and makes the approach resemble a controlled patching demonstration rather than a broadly applicable repair mechanism. Second, the method requires access to a full-precision reference model during inference, which is often impractical in real-world deployment scenarios where quantization is used to reduce cost and memory. Third, the empirical evidence is narrow and weak: experiments are limited to a single model family and quantization setting, with only 10 curated samples analyzed qualitatively. Among these, only two cases showed clear recovery, a modest success rate that does not support the claim that quantization-induced failures are highly localized. Finally, the stimulation strength α requires manual tuning and exhibits non-monotonic effects, further undermining the robustness and interpretability of the approach.

Overall Assessment: 2 = Resubmit next cycle: I think this paper needs substantial revisions that can be completed by the next ARR cycle.

Reported Issues:  No

Publication Ethics Policy Compliance: I did not use any generative AI tools for this review

Add: **Author-Editor Confidential Comment**

Official Review of Submission7892 by Reviewer oL3r

Official Review by Reviewer oL3r  08 Feb 2026, 05:31 (modified: 19 Feb 2026, 13:32)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer oL3r

 Revisions (/revisions?id=9yOnz6U7G0)

Paper Summary:

This paper studies a practically important failure mode: noticeably degrading in a few instruction-following examples. The authors propose Deep Attention Stimulation (DAS), a training-free, head-level intervention. DAS first diagnoses "degraded heads" by comparing attention behaviors between an FP16 reference model and a GPTQ 4-bit model, focusing on instruction-bearing tokens, and then manually repairs the quantized model by injecting a scaled correction signal into selected attention-head outputs at inference time (using a stimulation strength parameter α). Experiments are presented primarily on Qwen2.5-7B-Instruct under GPTQ 4-bit, using IFEval-derived failure cases, with a qualitative emphasis on a curated set of representative examples and analysis of non-monotonic α effects.

Summary Of Strengths:

- The paper addresses a timely and practically relevant question: quantization is central for deployment, yet its impact on instruction-following quality is often under-measured by standard metrics. The work makes this gap explicit and highlights concrete failure types that matter in real usage.
- The head-level adjustment "diagnose to intervene" framing is conceptually interesting. Rather than treating quantization damage as diffuse, the paper attempts a mechanistic angle by localizing perturbations to specific attention components and testing targeted interventions.

- The work explores a training-free post-hoc repair direction and openly reports instability regimes (e.g., over-stimulation leading to degeneration).

Summary Of Weaknesses:

1. Generalization is structurally unaddressed due to prompt-specific correction: The current setup caches and applies correction signals in a way that is effectively tied to the same prompts used for diagnosis. This makes the method look closer to a controlled patching demo than a generally applicable repair mechanism. Without evidence on unseen prompts (or a prompt-agnostic head profile), the central claim risks collapsing into *"we can fix these specific cases if we precompute prompt-conditioned deltas"*.
2. The approach appears to be dependant to FP16 model(teacher): The diagnosis and corrective signal construction rely on access to full-precision behavior as a reference. In real deployment contexts where quantization is pursued for cost, requiring an FP16 (or non-quantized) counterpart is a strong assumption that should be foregrounded and scoped. As written, the method can be interpreted as injecting teacher residuals rather than recovering an intrinsic capability of the quantized model.
3. α sensitivity and manual tuning undermine robustness and interpretability of gains. The paper itself shows non-monotonic behavior where the choice of α can flip outcomes (including inducing degeneration at higher values). Because α is selected via manual tuning on held-out failures, it becomes hard to disentangle "DAS works" from " α was chosen to make it work on this small evaluation slice", and it raises practical concerns about safe application without extensive calibration.
4. Empirical coverage is narrow and the evidence base is small: The main demonstration centers on a limited set of curated failures (e.g. a small qualitative sample) under a single model family and a single quantization bit setting. Given how quantization artifacts can vary by model, quantizer, calibration, and decoding configuration, the current evaluation does not support broad claims about the phenomenon or the proposed repair strategy's reliability.
5. I reviewed the bibliography due to recent concerns about problematic references, and I could not locate the cited works "A mechanistic interpretability analysis of grokking" or "Evaluating instruction following in language models" based on the current reference entries. Please double-check the validity of these citations.

Comments Suggestions And Typos:

Please see the weakness section.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness: 2 = Poor: Some of the main claims are not sufficiently supported. There are major technical/methodological problems.

Excitement: 2 = Potentially Interesting: this paper does not resonate with me, but it might with others in the *ACL community.

Overall Assessment: 2 = Resubmit next cycle: I think this paper needs substantial revisions that can be completed by the next ARR cycle.

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 5 = They could easily reproduce the results.

Datasets: 1 = No usable datasets submitted.

Software: 1 = No usable software released.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

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Official Review of Submission7892 by Reviewer CMo1

Official Review by Reviewer CMo1 06 Feb 2026, 08:18 (modified: 19 Feb 2026, 13:32)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer CMo1
Revisions (/revisions?id=3YkNuR1dt3)

Paper Summary:
This paper identifies an interesting issue in the deployment of large language models: the attention patterns of quantized LLMs may differ from those of their full-precision counterparts, leading to degradation in instruction-following performance. To address this, the authors propose a detection and mitigation framework.
For detection, they compute a difference score by comparing the averaged attention weights of each attention head between the FP4 and FP32 models, and then select the heads with the largest difference scores. For mitigation, they add steering vectors to the context hidden states to compensate for the identified discrepancies.

Summary Of Strengths:
The research problem is novel and interesting.

Summary Of Weaknesses:

1. The mitigation method is strictly constrained to limited inputs, which restricts its applicability to broader test scenarios.
2. The evaluation is mainly conducted on only 10 samples, making it difficult to assess generalization to more realistic settings.
3. Experiments are limited to Qwen2.5-7B and GPTQ quantization. Evaluating additional models and quantization methods would strengthen the empirical evidence and overall impact of the work.

Comments Suggestions And Typos:
N/A

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.
Soundness: 2.5
Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.
Overall Assessment: 2 = Resubmit next cycle: I think this paper needs substantial revisions that can be completed by the next ARR cycle.
Ethical Concerns:
There are no concerns with this submission

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.
Datasets: 1 = No usable datasets submitted.
Software: 1 = No usable software released.

Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

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Official Review of Submission7892 by Reviewer VsXW

Official Review by Reviewer VsXW 📅 05 Feb 2026, 01:19 (modified: 19 Feb 2026, 13:32)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer VsXW

📄 Revisions (/revisions?id=cB9w1Y5Hlj)

Paper Summary:

This paper investigates the degradation of instruction-following abilities in quantized LLMs. The aim is to determine whether this performance drop stems from global model degradation or is localized within specific internal components. To this end, the authors propose Deep Attention Stimulation (DAS), a diagnostic intervention that injects a correction signal into selected attention heads. The signal is computed as the difference between the full-precision and quantized attention outputs. The authors claim that this intervention recovers instruction-following abilities, leading them to conclude that quantization-induced failures in instruction following are highly localized in a small subset of attention heads.

Summary Of Strengths:

- Probing the mechanistic causes of quantization-induced performance drops is a timely and interesting direction for model efficiency research.
- The manuscript is well-structured, and the methodology is easy to follow.

Summary Of Weaknesses:

- The concept of "patching" or "intervening" on neural components with activations from a reference model to perform causal mediation analysis is not new; Similar approaches have been pioneered in works such as [1]. To establish DAS as a standalone contribution, the authors should explicitly contrast it with existing causal analysis methods [1,2,3] and demonstrate its unique advantages.
- The experiment results do not seem to justify the authors' conclusions. Specifically, the report indicates that instruction-following was recovered in only 2 out of 10 samples. This sample size is too small for a robust claim, and the 20% success rate is modest. If 80% of the cases fail to recover via DAS, it suggests that the degradation may indeed be a global phenomenon rather than a localized one. The authors need to address why the majority of cases remain uncorrected by their method.

[1] Kevin Meng, David Bau, Alex Andonian, Yonatan Belinkov. Locating and Editing Factual Associations in GPT. In NeurIPS 2022. [2] Alessandro Stolfo, Yonatan Belinkov, and Mrinmaya Sachan. 2023. A Mechanistic Interpretation of Arithmetic Reasoning in Language Models using Causal Mediation Analysis. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 7035–7052, Singapore. Association for Computational Linguistics. [3] Yaniv Nikankin, Anja Reusch, Aaron Mueller, Yonatan Belinkov. Arithmetic Without Algorithms: Language Models Solve Math with a Bag of Heuristics. In ICLR 2025.

Comments Suggestions And Typos:

- In Figure 4 the caption writes "The right panel..." while the right panel is absent.
- I suggest the authors expand their evaluation beyond 10 samples to provide a statistically significant view of the "localization" v.s. "global degradation" debate.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness: 2 = Poor: Some of the main claims are not sufficiently supported. There are major technical/methodological problems.

Excitement: 2 = Potentially Interesting: this paper does not resonate with me, but it might with others in the *ACL community.

Overall Assessment: 2 = Resubmit next cycle: I think this paper needs substantial revisions that can be completed by the next ARR cycle.

Ethical Concerns:

There are no concerns with this submission

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 1 = No usable datasets submitted.

Software: 1 = No usable software released.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

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